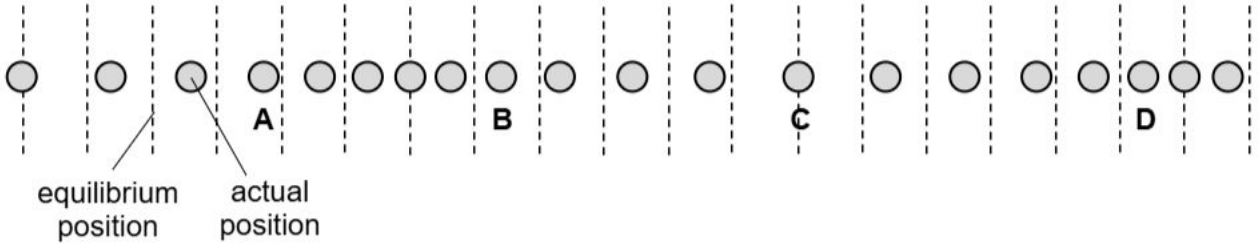
**14** A longitudinal sound wave travels from left to right through air.

The diagram below shows the equilibrium positions and the actual positions of a series of air particles through which the sound wave is passing.




Which of the particles has an acceleration that is opposite in direction to its velocity at this instant?

Ans: **D**

Hint: sketch the displacement-distance graph of the particles.

A: The particle is at its amplitude position, so there is no velocity (velocity is zero) at this instant.

B: The particle is moving towards equilibrium position and velocity is in the same direction as acceleration directed towards equilibrium position.

C: The particle is at the equilibrium position and has maximum velocity but zero acceleration.

D: The particle is moving away from equilibrium position and velocity is in the opposite direction as acceleration directed towards equilibrium position.

**15** Light waves from a single slit pass through a double-slit and interfere. The curved lines represent